```
> restart;
> # Problem 1: evaluate the number using floating point arithmetic
> evalf(1/2);
```

$$0.5000000000 \tag{1}$$

```
> evalf(exp(1));
```

$$2.718281828 \tag{2}$$

```
> exp(1);
```

$$\mathrm{e} \tag{3}$$

```
> evalf(sqrt(3));
```

$$1.732050808 \tag{4}$$

```
> sqrt(3);
```

$$\sqrt{3} \tag{5}$$

```
> pi
```

$$\pi \tag{6}$$

```
> Pi
```

$$\pi \tag{7}$$

```
> evalf(Pi);
```

$$3.141592654 \tag{8}$$

```
> evalf(pi);
```

$$\pi \tag{9}$$

```
> # Problem 2: assign the expression to a variable and then expand
  it
> # a)
> expression:=(x^2+2*x-1)*(x^2-2);
```

$$expression := (x^2 + 2\,x - 1)\,(x^2 - 2) \tag{10}$$

```
> # b)
> expression:="expression";
```

$$expression := \text{"expression"} \tag{11}$$

```
> expression
```

$$\text{"expression"} \tag{12}$$

```
> expression:='expression';
```

$$expression := expression \tag{13}$$

```
> expression
```

$$expression \tag{14}$$

```
> expression:=(x+n)^5;
```

$$expression := (x + n)^5 \tag{15}$$

```
> expression:='expression';
```

$$expression := expression \tag{16}$$

```
> # Problem 3: factorize x^8-1
> ec1:=x^8-1;
```

$$ec1 := x^8 - 1 \quad (17)$$

```
> ec2:=factor(ec1);
```

$$ec2 := (x-1)(x+1)(x^2+1)(x^4+1) \quad (18)$$

```
> is(ec1=ec2);
```

$$true \quad (19)$$

```
> restart;
> # Problem 4: add the rational expression by applying the factor command
> factor(2*x^2/(x^3-1)+3*x/(x^2-1));
```

$$\frac{(5x^2+5x+3)x}{(x-1)(x+1)(x^2+x+1)} \quad (20)$$

```
> simplify(expression);
```

$$expression \quad (21)$$

```
> restart;
> # Problem 5: simplify the trigonometric expression using simplify (expression, trig)
> expression:=(sin(x))^2+(cos(x))^2;
```

$$expression := \sin(x)^2 + \cos(x)^2 \quad (22)$$

```
> simplify(expression,trig);
```

$$1 \quad (23)$$

```
> simplify(2/4);
```

$$\frac{1}{2} \quad (24)$$

```
> restart;
> # Problem 6: evaluate using subs and eval the expression
> subs(x=1,exp(x)+ln(x));
```

$$e + \ln(1) \quad (25)$$

```
> eval(exp(x)+ln(x), x=1);
```

$$e \quad (26)$$

```
> restart;
> # Problem 7: solve
> #  a)
> solve(x^2-4*x+3=0,x);
```

$$3, 1 \quad (27)$$

```
> #  b)
> solve(x^2*y+2*y-x=0,y);
```

$$\frac{x}{x^2+2} \quad (28)$$

```
> #  c)
> solve(x^2*y+2*y-x=0,x);
```

(29)

$$\frac{1+\sqrt{-8y^2+1}}{2y}, -\frac{-1+\sqrt{-8y^2+1}}{2y} \quad (29)$$

```
> # d)
> solve(x-cos(x)=0, x);
```

$$RootOf(\_Z-\cos(\_Z)) \quad (30)$$

```
> fsolve(x-cos(x)=0,x);
```

$$0.7390851332 \quad (31)$$

```
> # e)
> fsolve(x^5-3*x^3-1=0,x);
```

$$-1.668777593, -0.7418139305, 1.782308780 \quad (32)$$

```
> # f)
> ec1:=4*x+3*y=10;
```

$$ec1 := 4x+3y=10 \quad (33)$$

```
> ec2:=3*x-y=1;
```

$$ec2 := 3x-y=1 \quad (34)$$

```
> ec1:='ec1';
```

$$ec1 := ec1 \quad (35)$$

```
> ec1;
```

$$ec1 \quad (36)$$

```
> ec2:='ec2';
```

$$ec2 := ec2 \quad (37)$$

```
> ec2
```

$$ec2 \quad (38)$$

```
> eq1:=4*x+3*y=10;
```

$$eq1 := 4x+3y=10 \quad (39)$$

```
> eq2:=3*x-y=1;
```

$$eq2 := 3x-y=1 \quad (40)$$

```
> solve({eq1,eq2},{x,y});
```

$$\{x=1, y=2\} \quad (41)$$

```
> restart;
> # Problem 8: assign to a variable f the function f, evaluate f
  (0), f(1), f(-1), calculate first and second order derivatives of
  f, a primitive of f, evaluate integral from -1 to 1 of f,
  unnasign f, derivative in 0
> # 1st correct way:
> f:=x->exp(x)-sin(x);
```

$$f := x \mapsto e^x - \sin(x) \quad (42)$$

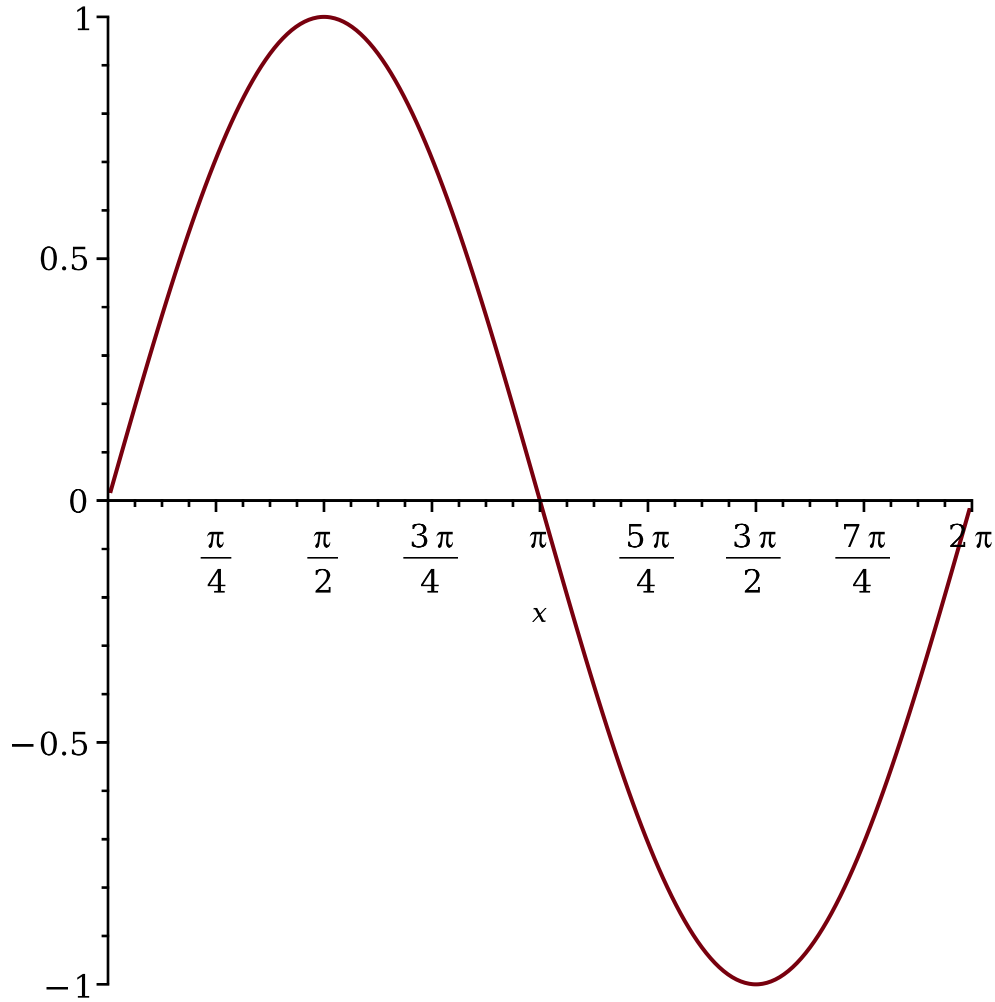
```
> f(x)
```

$$e^x - \sin(x) \quad (43)$$

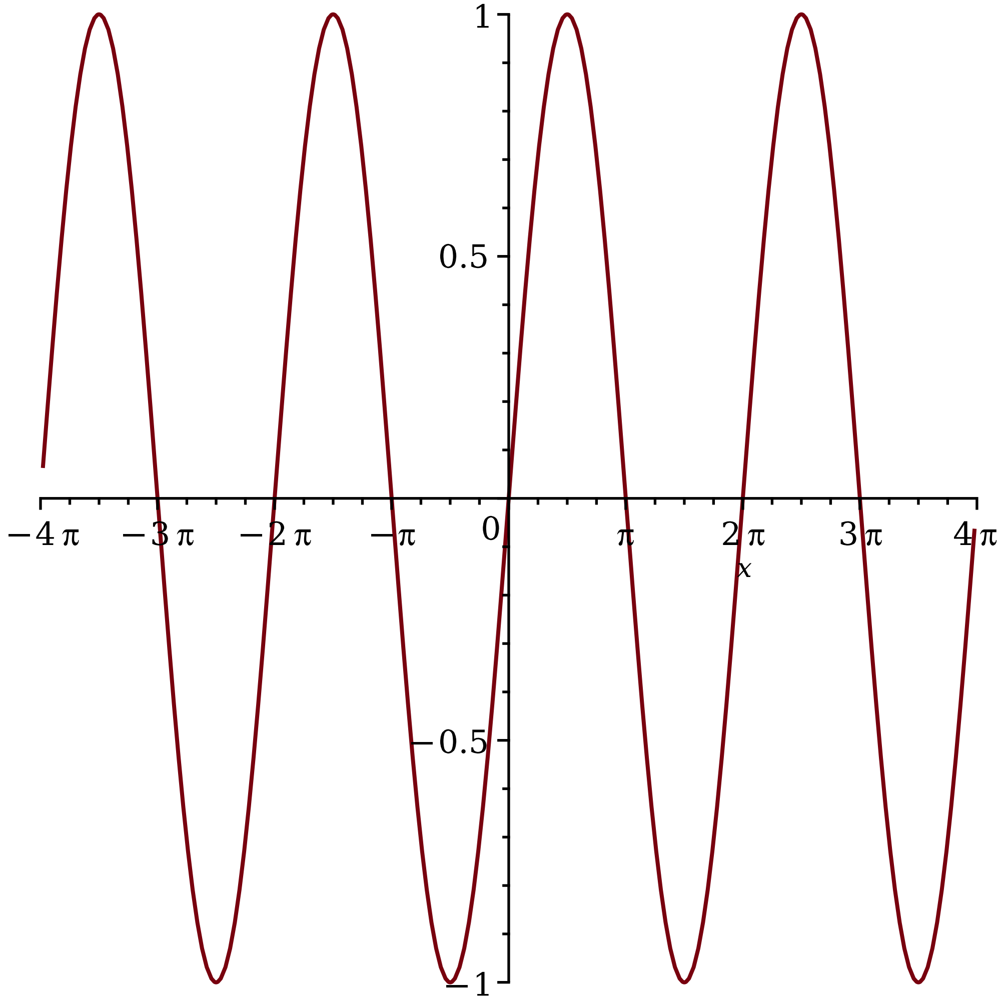
```
> # 2nd correct way:
```

```
> f:=unapply(exp(x)-sin(x), x);
```
$$f := x \mapsto e^{x} - \sin(x) \tag{44}$$
```
> f(x)
```
$$e^{x} - \sin(x) \tag{45}$$
```
> f(0);f(1);f(-1);
```
$$1$$
$$e - \sin(1)$$
$$e^{-1} + \sin(1) \tag{46}$$
```
> # 1st order derivative using D:
> D(f)(x);
```
$$e^{x} - \cos(x) \tag{47}$$
```
> # 2nd order derivative using D:
> (D@@2)(f)(x);
```
$$e^{x} + \sin(x) \tag{48}$$
```
> # 1st order derivative using diff:
> diff(f(x),x);
```
$$e^{x} - \cos(x) \tag{49}$$
```
> # 2nd order derivative using diff:
> diff(f(x),x$2);
```
$$e^{x} + \sin(x) \tag{50}$$
```
> # primitive of f
> int(f(x), x);
```
$$\cos(x) + e^{x} \tag{51}$$
```
> # integral
> int(f(x),x=-1..1);
```
$$-e^{-1} + e \tag{52}$$
```
> f(x):='f(x)';
```
$$f(x) := f(x) \tag{53}$$
```
> f(x);
```
$$f(x) \tag{54}$$
```
> # derivative in a given point (x=0):
> D(f)(0);
```
$$0 \tag{55}$$
```
> restart;
> # Problem 9: assign to variable g the expression, evaluate
  expression in x=0, compute first order derivative, evaluate it in
  x=0, find a primitive, evaluate second order derivative, evaluate
  it in x=0, evaluate integral from x=-1 to x=1 of g
> g:=exp(x)-sin(x);
```
$$g := e^{x} - \sin(x) \tag{56}$$

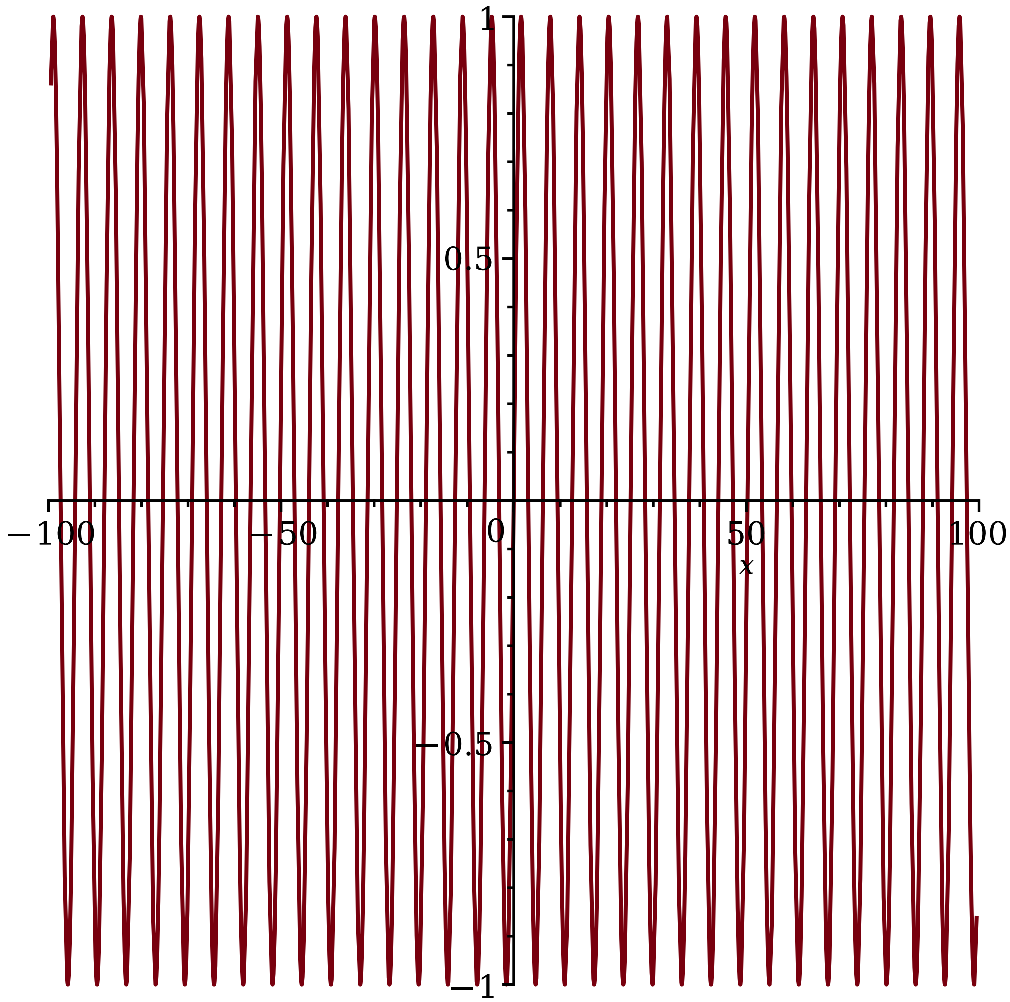
```
> eval(g, x=0);
```
$$1 \qquad (57)$$
```
> ec:=diff(g,x);
```
$$ec := e^{x} - \cos(x) \qquad (58)$$
```
> eval(ec, x=0);
```
$$0 \qquad (59)$$
```
> int(g,x);
```
$$\cos(x) + e^{x} \qquad (60)$$
```
> ecc:=diff(g,x$2);
```
$$ecc := e^{x} + \sin(x) \qquad (61)$$
```
> eval(ecc, x=0);
```
$$1 \qquad (62)$$
```
> int(g, x=-1..1);
```
$$-e^{-1} + e \qquad (63)$$
```
> restart;
> # Problem 10: limits
> limit(sin(x)/x,x=0);
```
$$1 \qquad (64)$$
```
> limit((cos(x)+1)/(x-Pi),x=Pi);
```
$$0 \qquad (65)$$
```
> restart;
> # Problem 11: plot the graph on given intervals
> plot(sin(x), x=0..2*Pi);
```



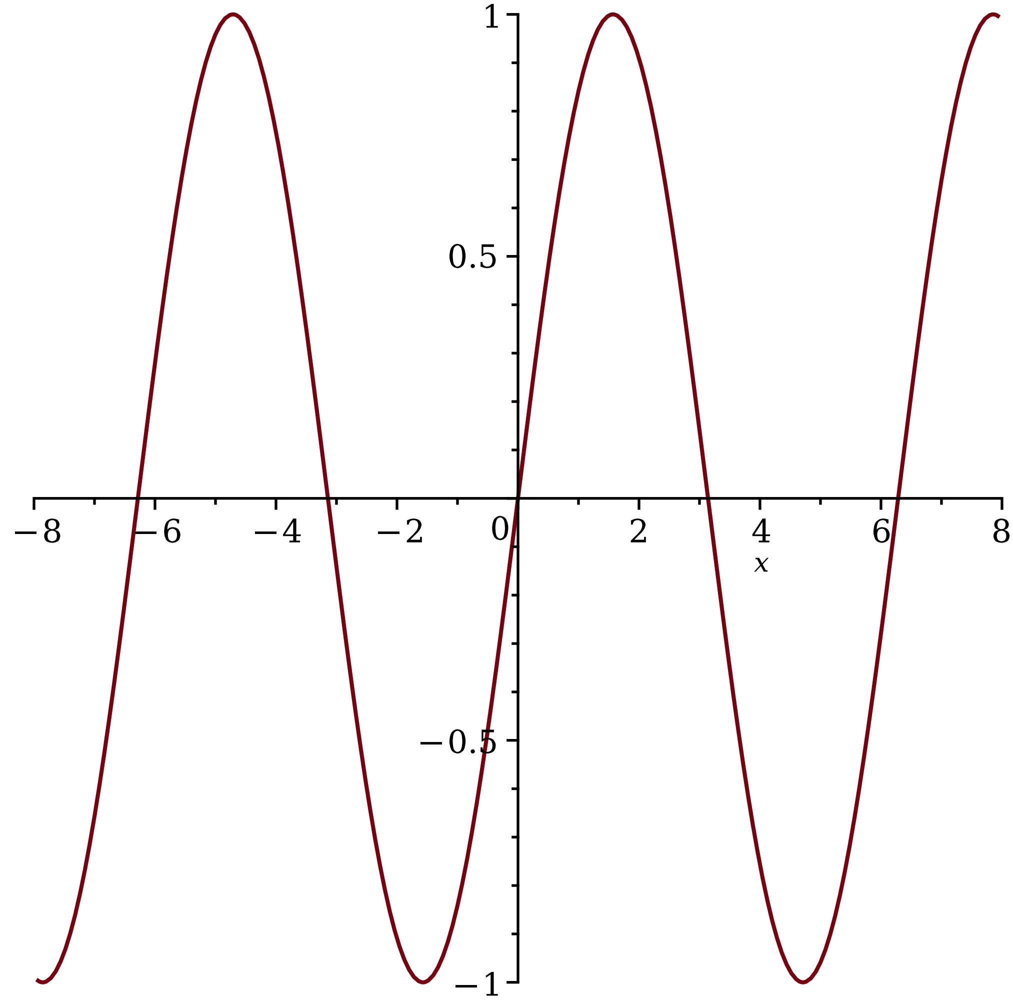

```
> plot(sin(x),x=-4*Pi..4*Pi);
```



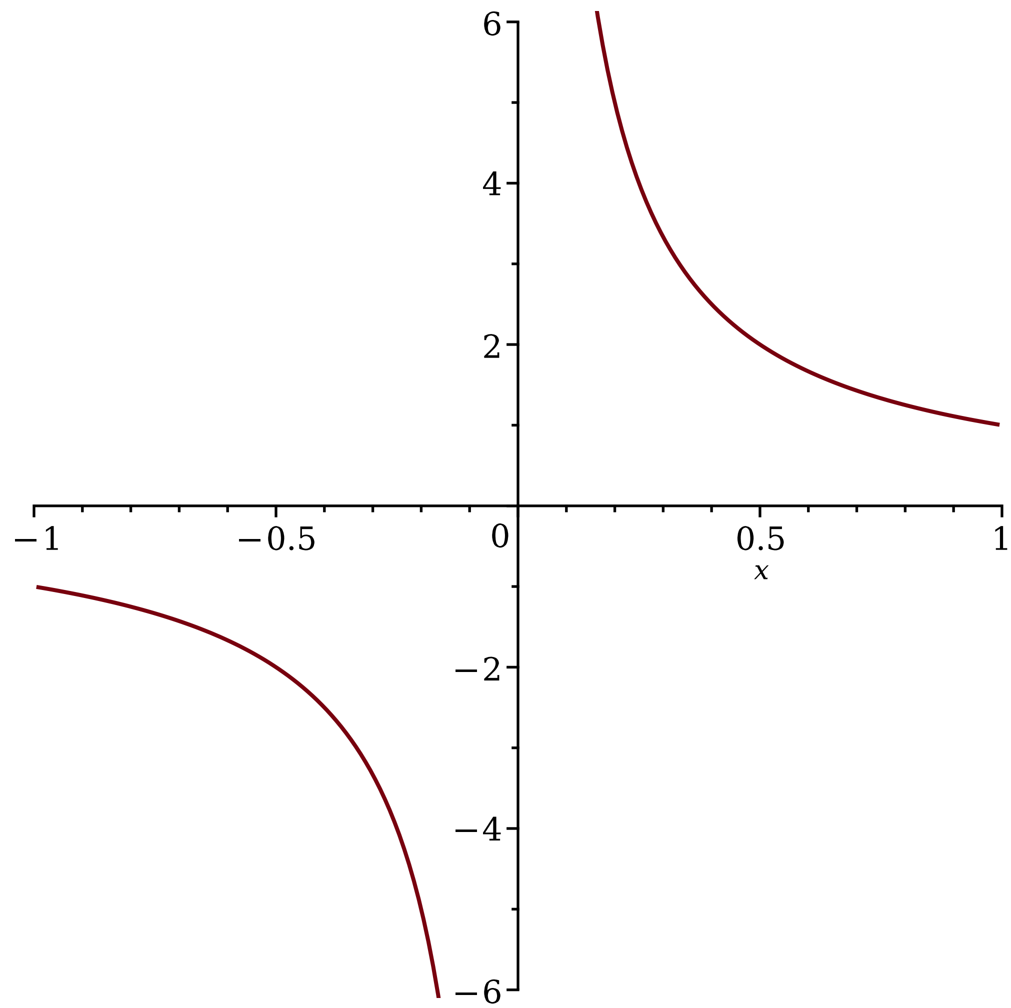

```
> plot(sin(x), x=-100..100);
```



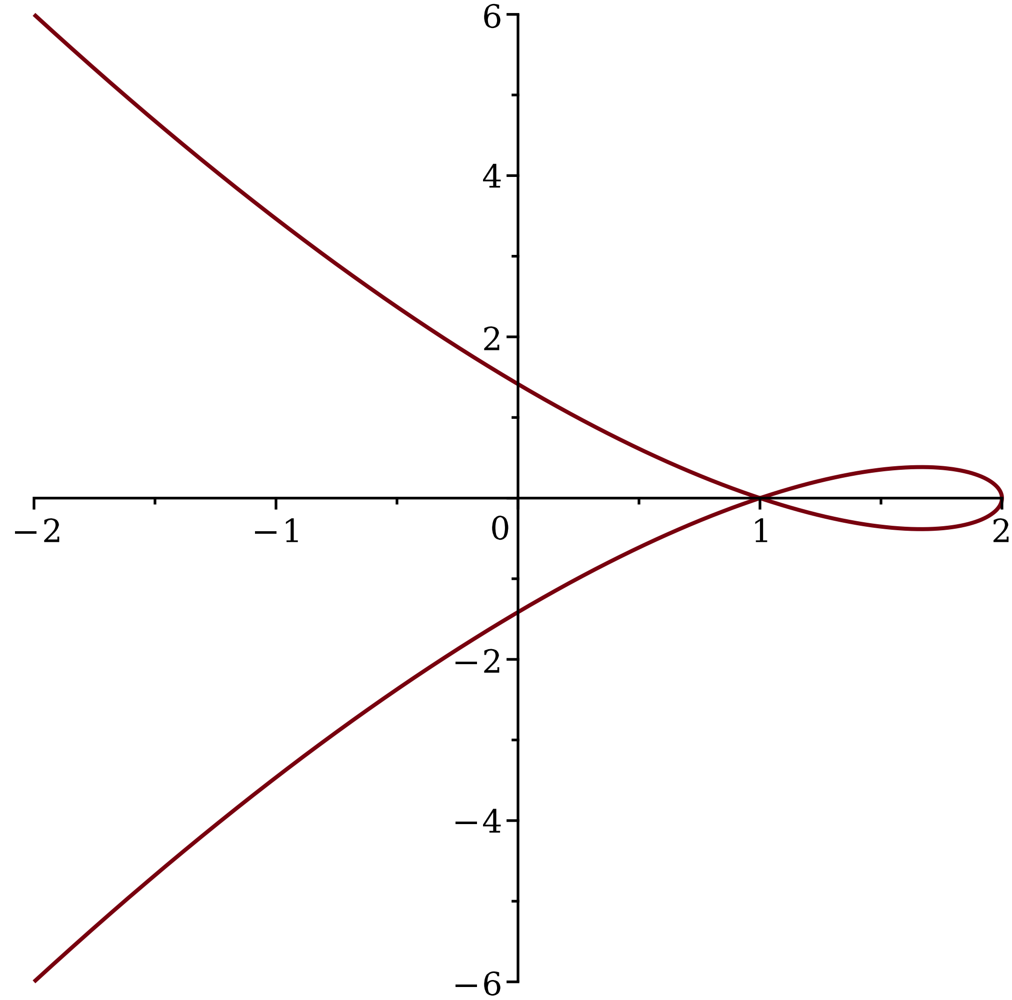

```
> plot(sin(x), x=-8..8);
```



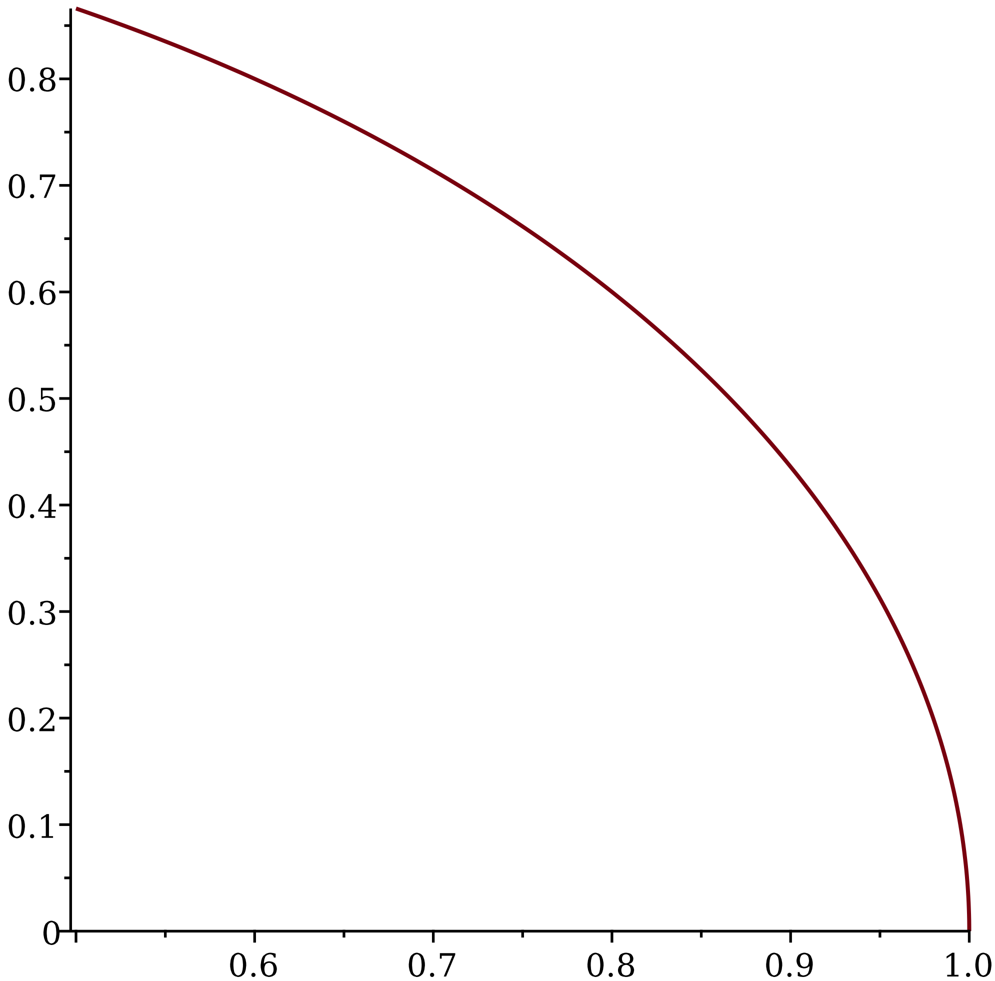

```
> restart;
> # Problem 12: plot the graph on given interval
> plot(1/x, x=-1..1);
```



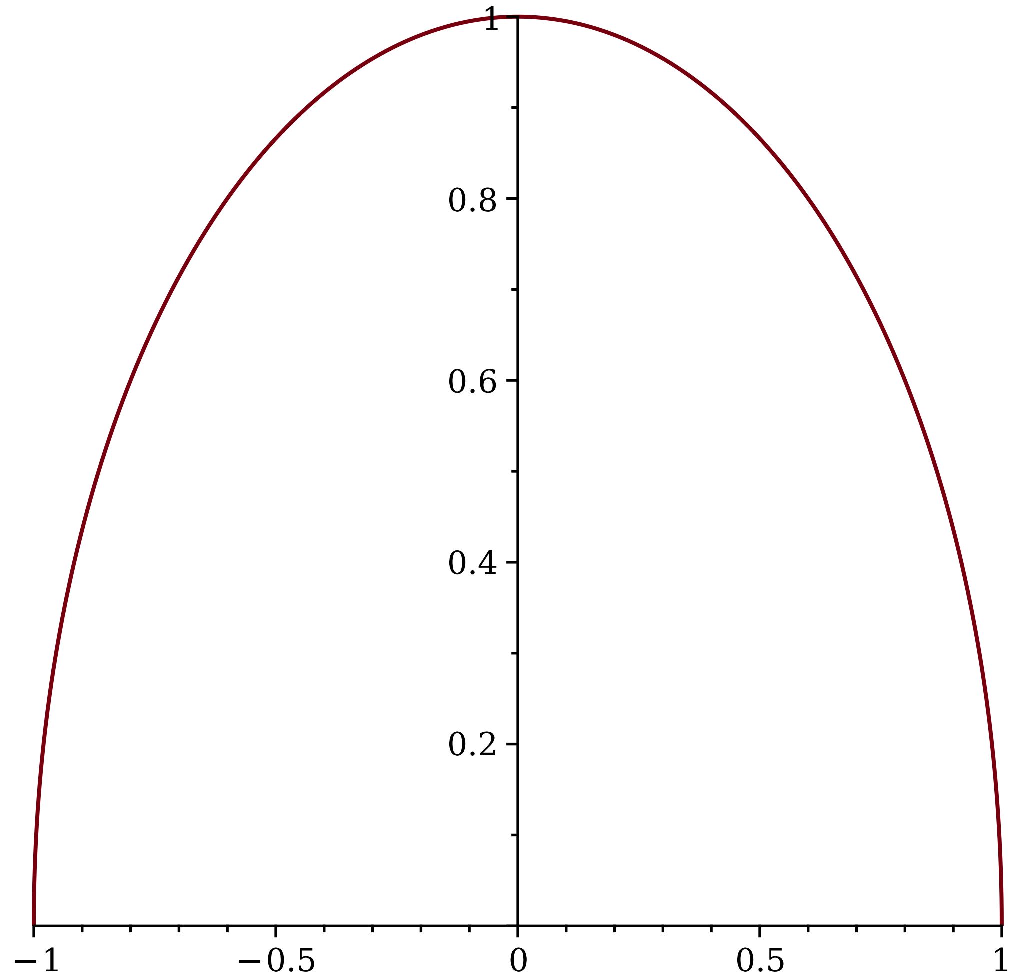

```
> restart;
> # Problem 13: plot the planar curve of parametric equations
> plot([2-t^2,t-t^3, t=-2..2]);
```



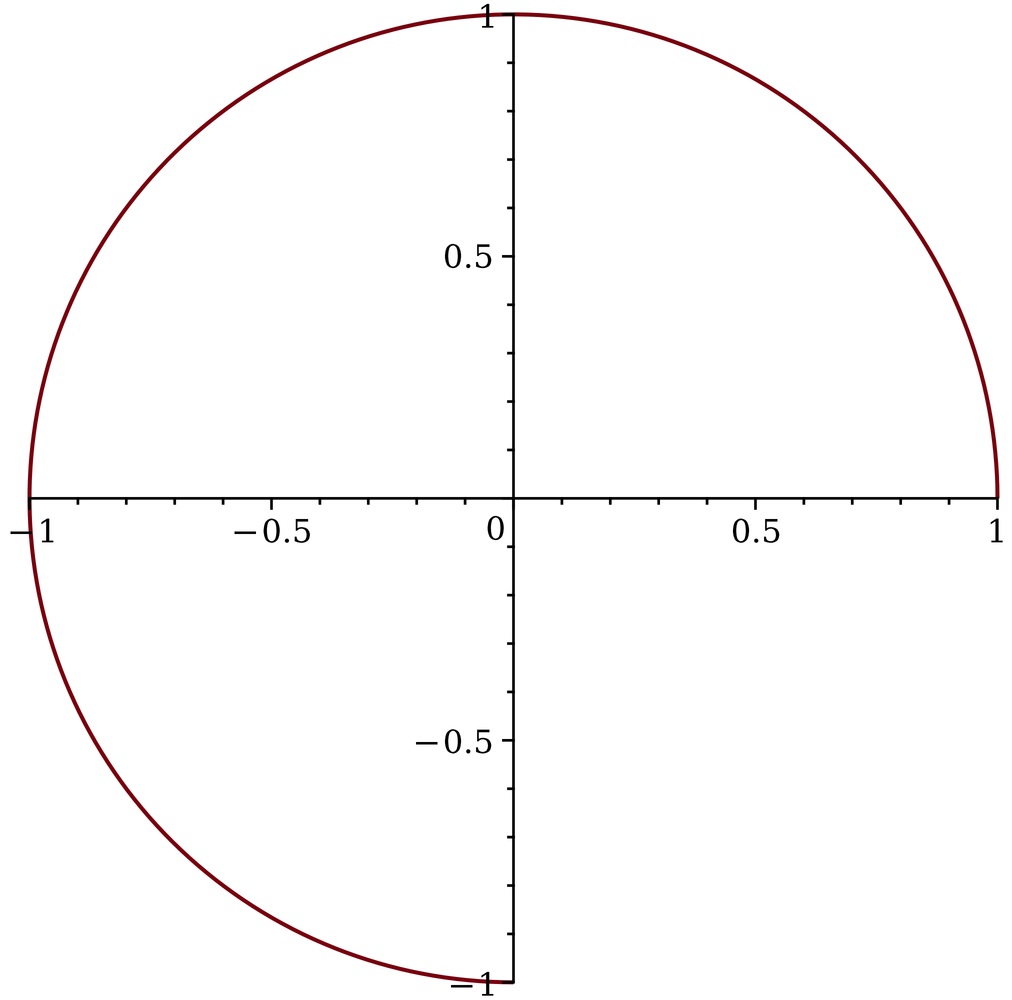

```
> restart;
> # Problem 14: planar curve of parametric equations
> plot([cos(t),sin(t), t=0..Pi/3]);
```



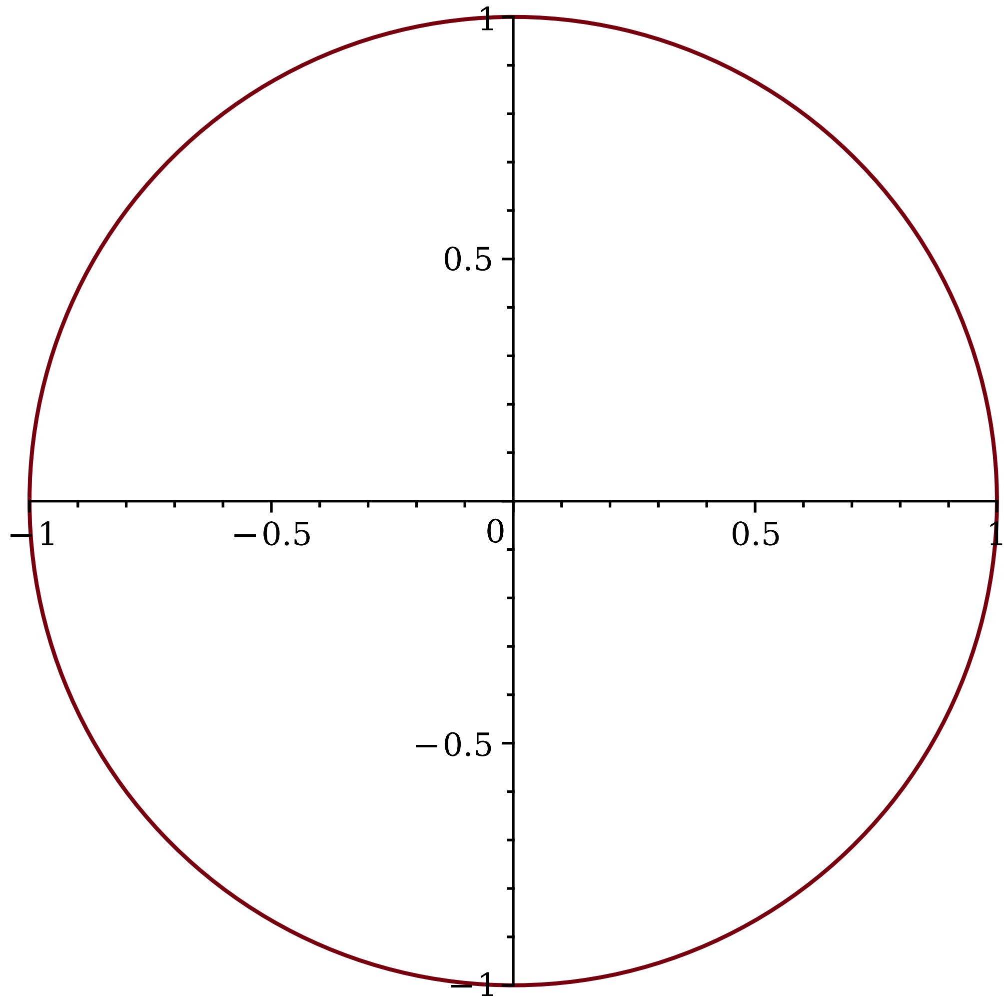

```
> plot([cos(t), sin(t), t=0..Pi]);
```



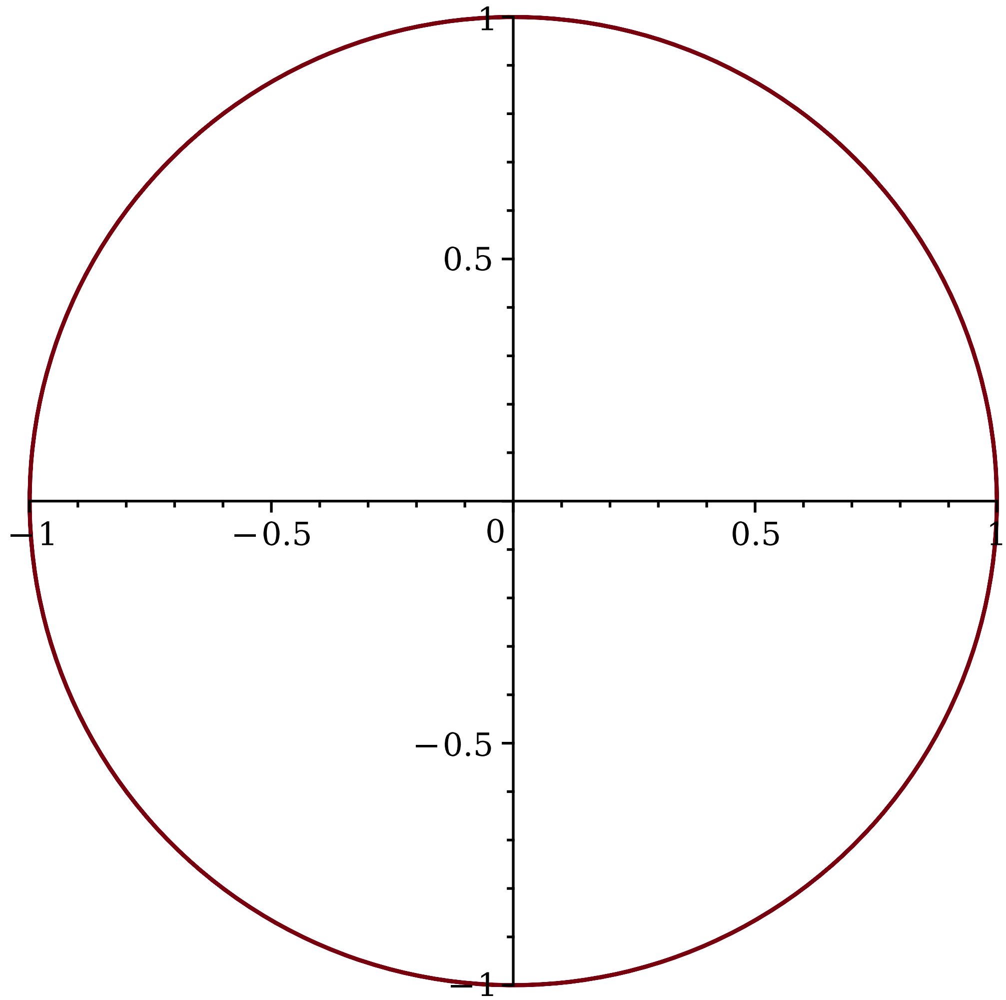

```
> plot([cos(t), sin(t), t=0..3*Pi/2]);
```



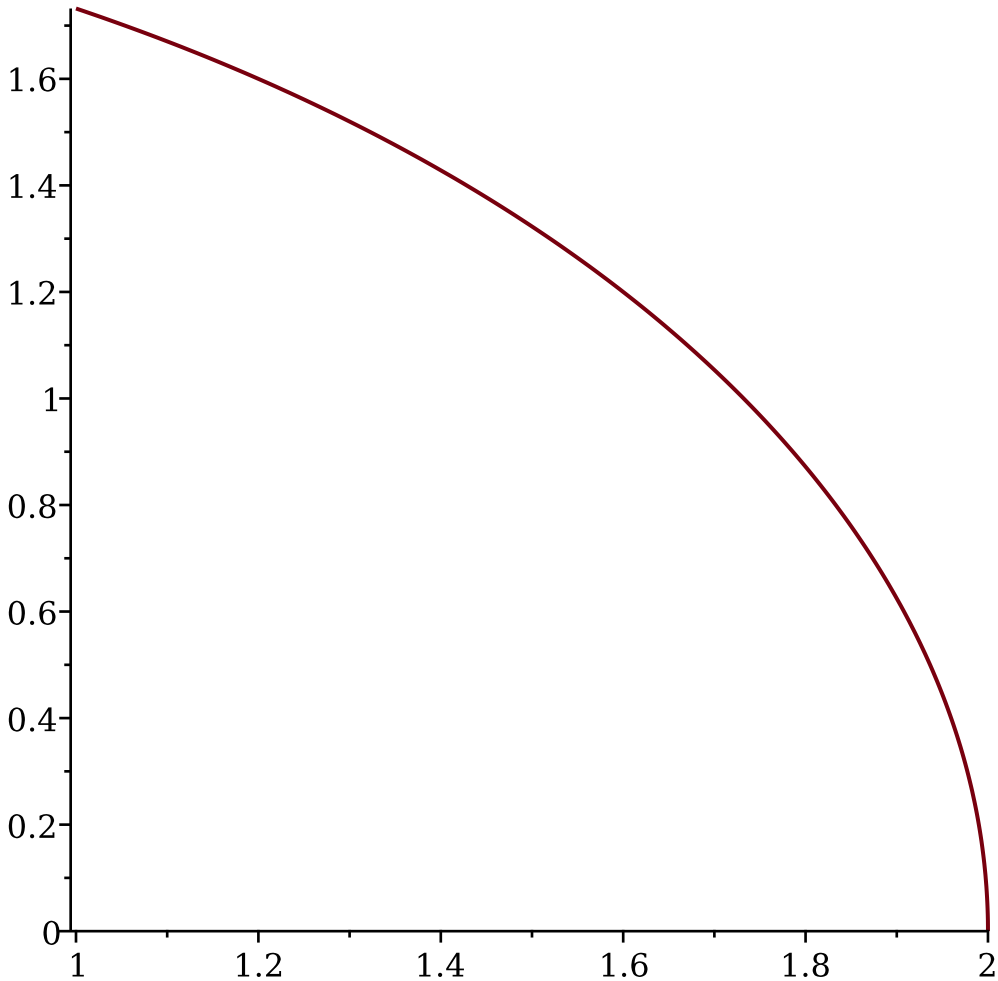

```
> plot([cos(t), sin(t), t=0..2*Pi]);
```



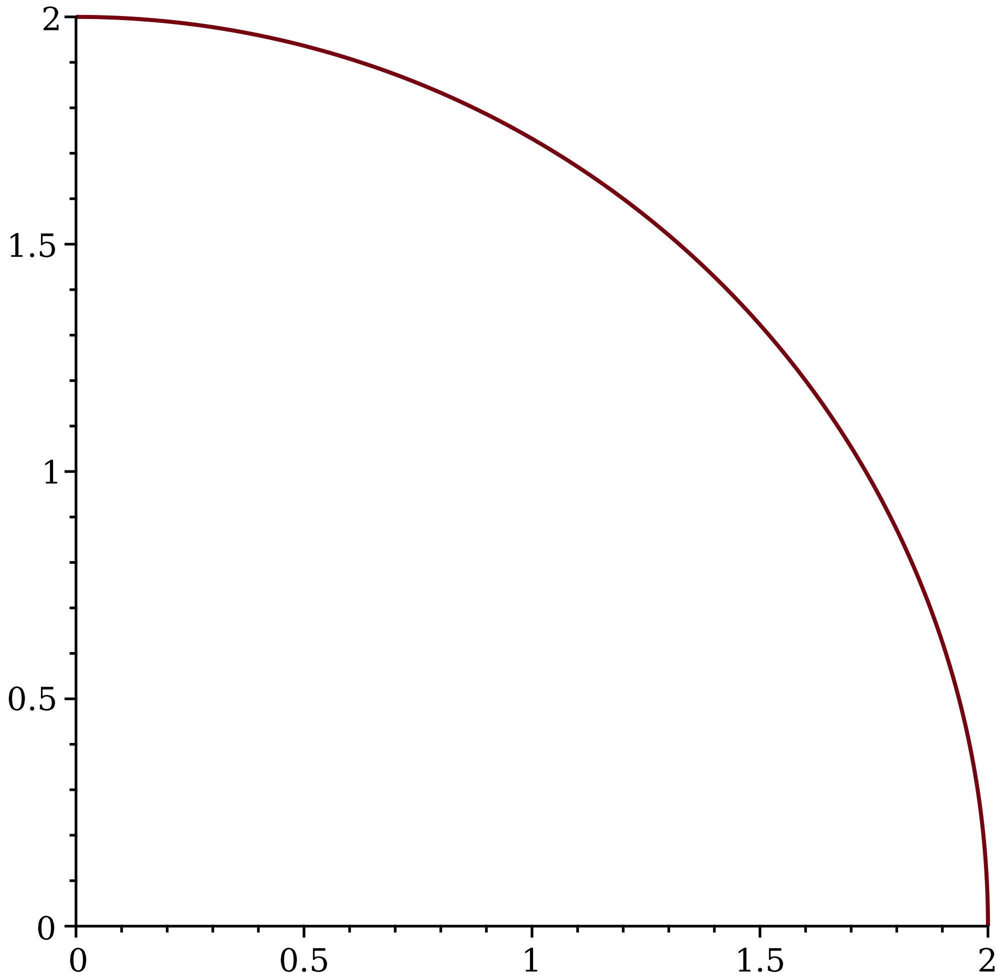

```
> plot([cos(t), sin(t), t=0..4*Pi]);
```



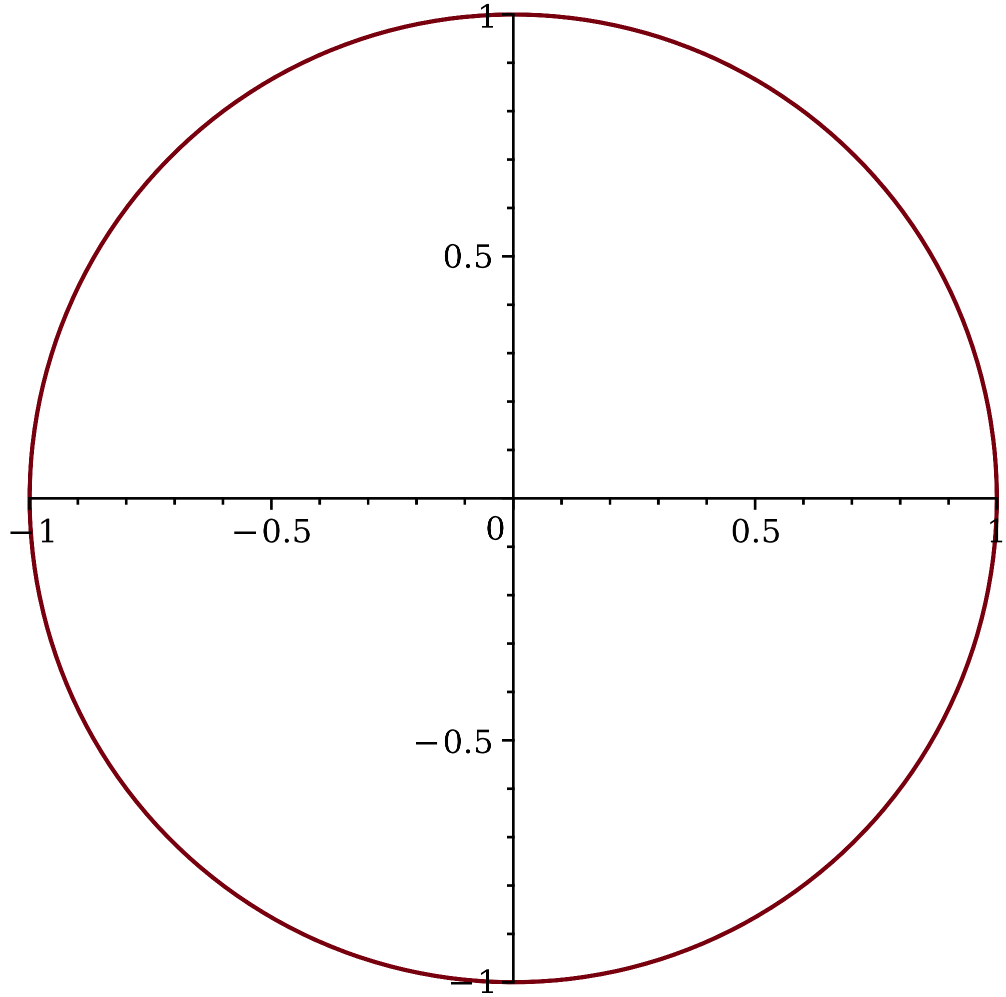

```
> restart;
> # Problem 15: plot thr planar curves of parametric equations
> # a)
> plot([2*cos(t/3),2*sin(t/3), t=0..Pi]);
```



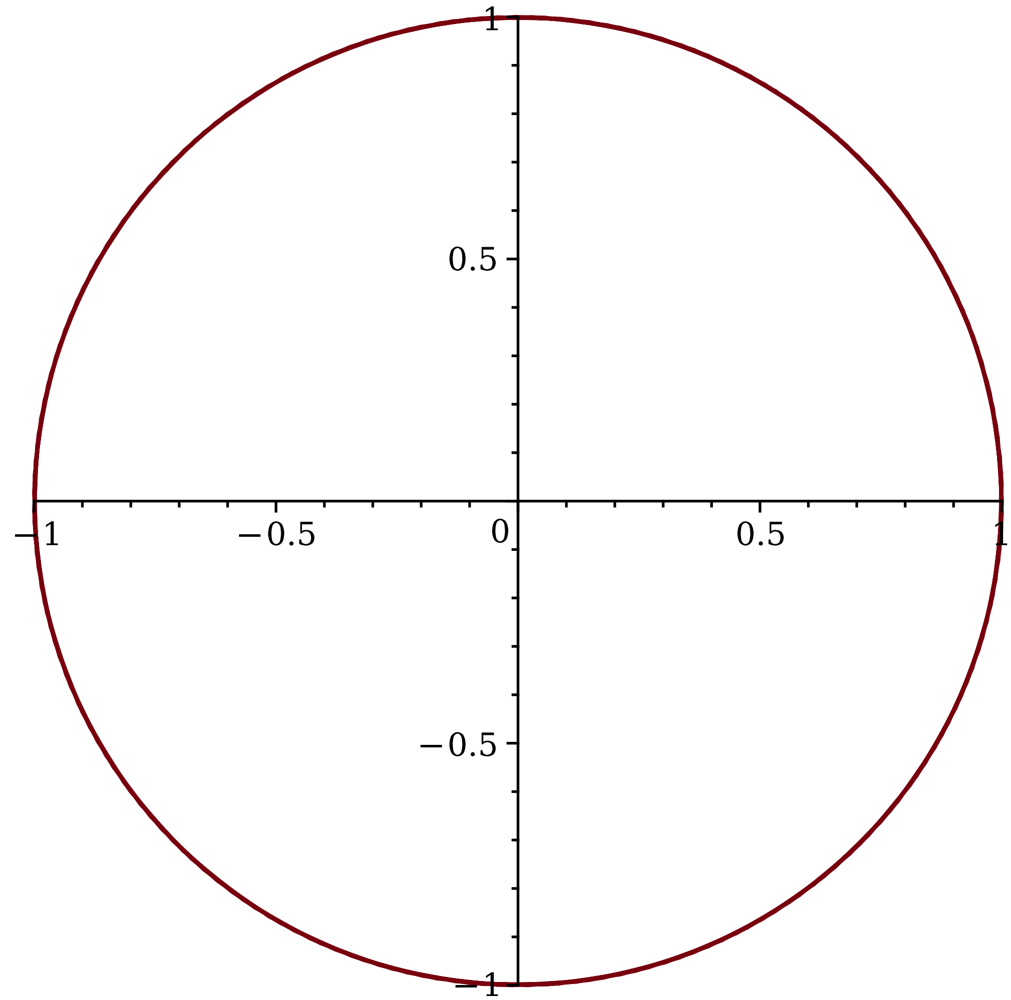

```
> plot([2*cos(t/3), 2*sin(t/3), t=0..3*Pi/2]);
```



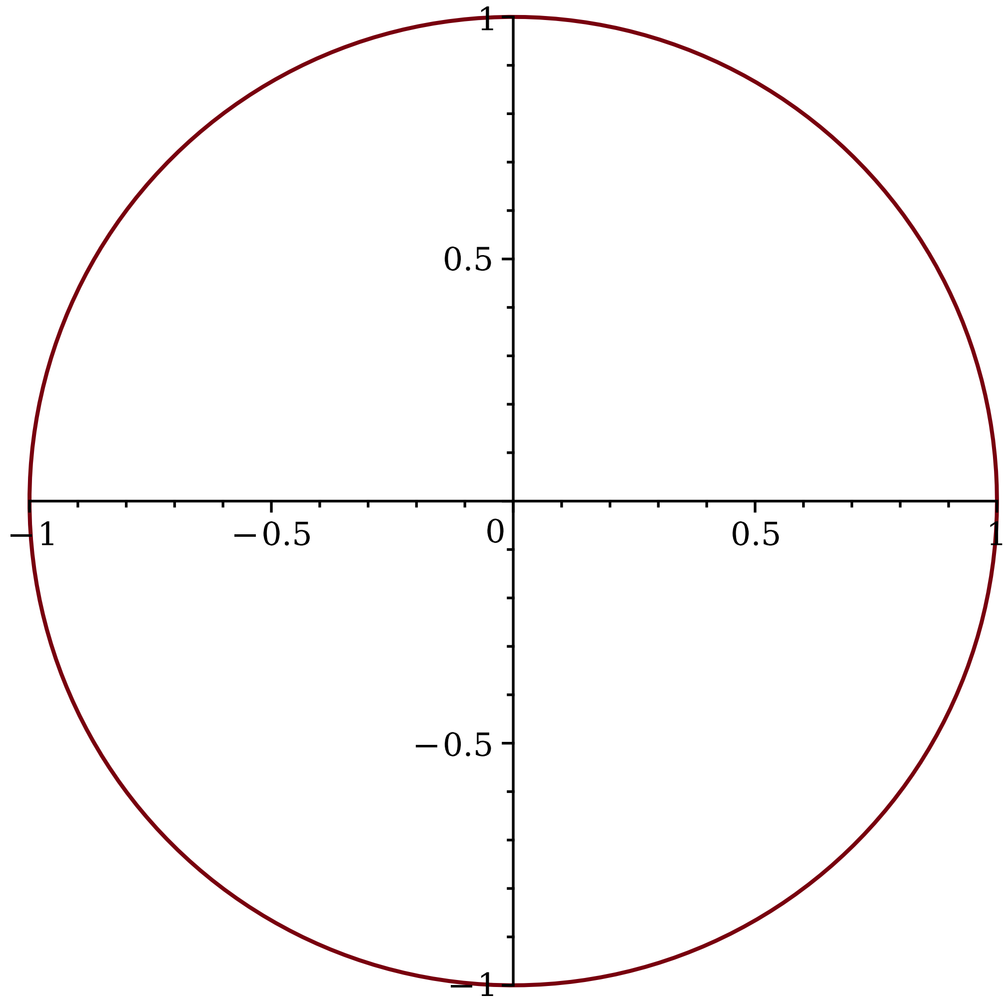

```
> # b)
> plot([cos(4*t), sin(4*t), t=0..Pi]);
```




```
> plot([cos(4*t), sin(4*t), t=0..2*Pi]);
```



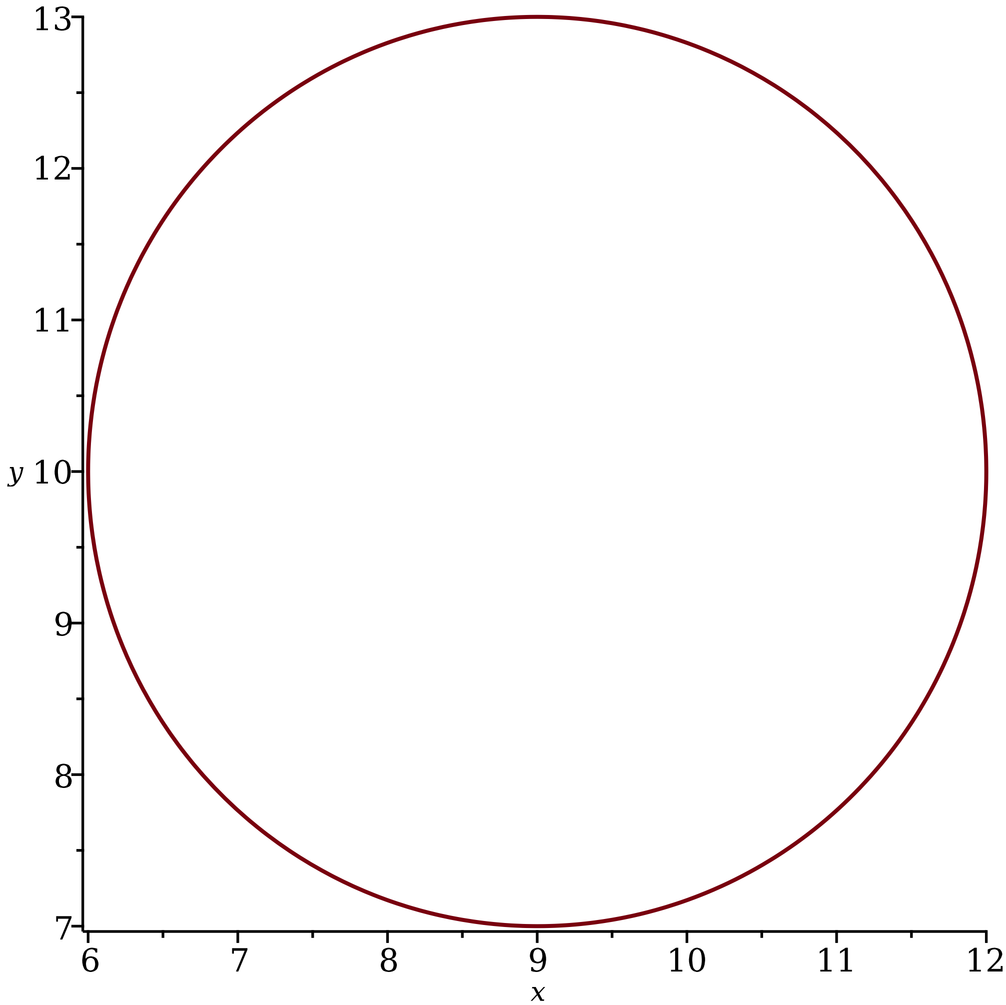

```
> plot([cos(4*t), sin(4*t), t=0..Pi/2]);
```



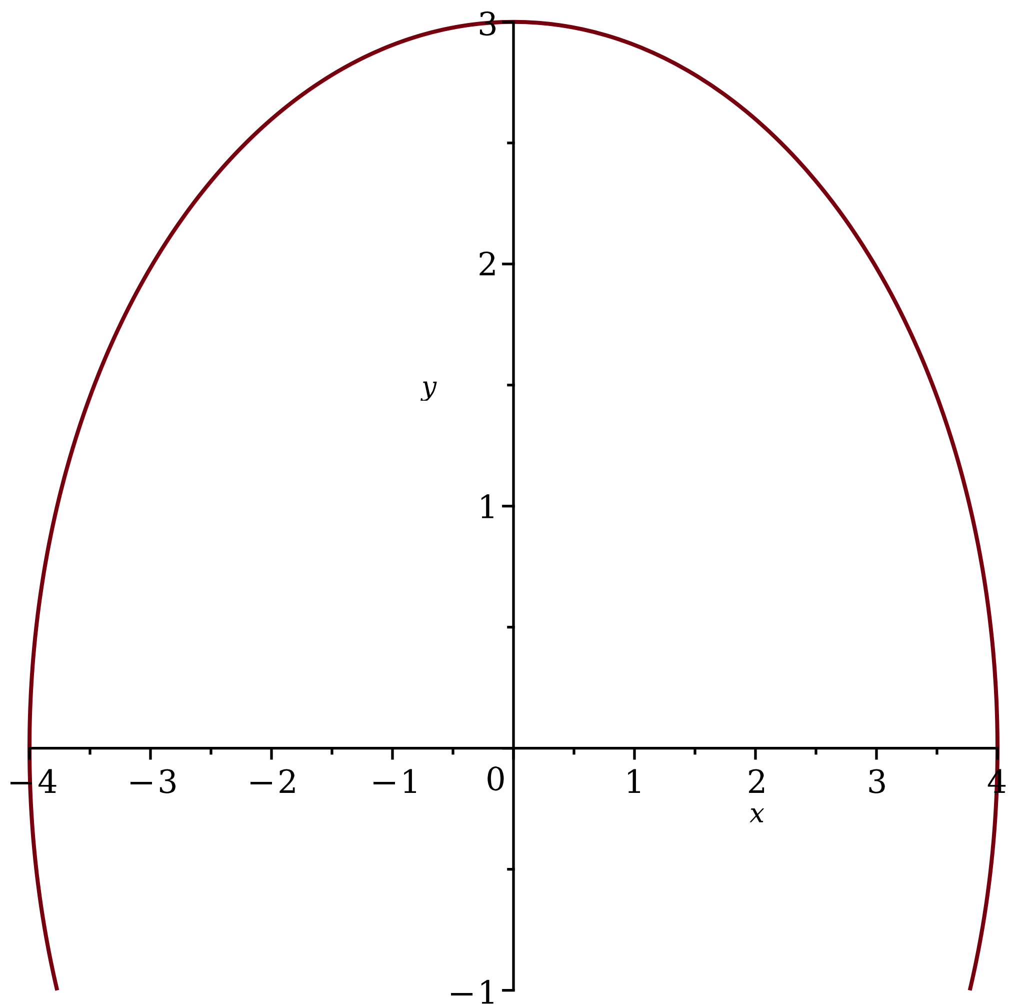

```
> restart;
> # Problem 16: implicit equations of a circle centered in (9,10),
  an ellipse and a parabola, plot at least a circle, an ellipse and
  a parabola
> with(plots);
```

[*animate, animate3d, animatecurve, arrow, changecoords, complexplot, complexplot3d, conformal, conformal3d, contourplot, contourplot3d, coordplot, coordplot3d, densityplot, display, dualaxisplot, fieldplot, fieldplot3d, gradplot, gradplot3d, implicitplot, implicitplot3d, inequal, interactive, interactiveparams, intersectplot, listcontplot, listcontplot3d, listdensityplot, listplot, listplot3d, loglogplot, logplot, matrixplot, multiple, odeplot, pareto, plotcompare, pointplot, pointplot3d, polarplot, polygonplot, polygonplot3d, polyhedra_supported, polyhedraplot, rootlocus, semilogplot, setcolors, setoptions, setoptions3d, shadebetween, spacecurve, sparsematrixplot, surfdata, textplot, textplot3d, tubeplot*] **(66)**

```
> implicitplot((x-9)^2+(y-10)^2=9, x=4..14, y=5..15);
```



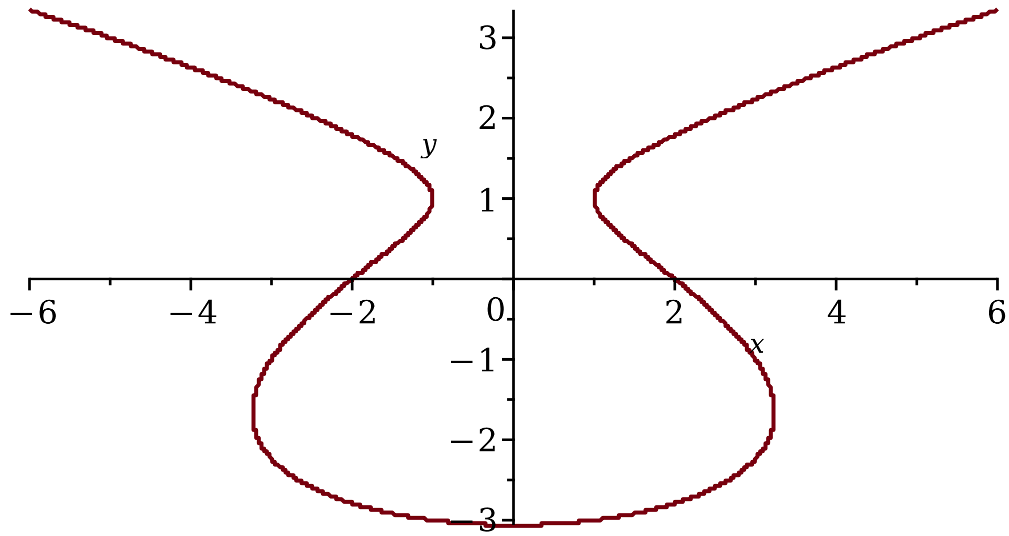

```
> implicitplot(x^2/16+y^2/9=1,x=-5..5, y=-1..25);
```



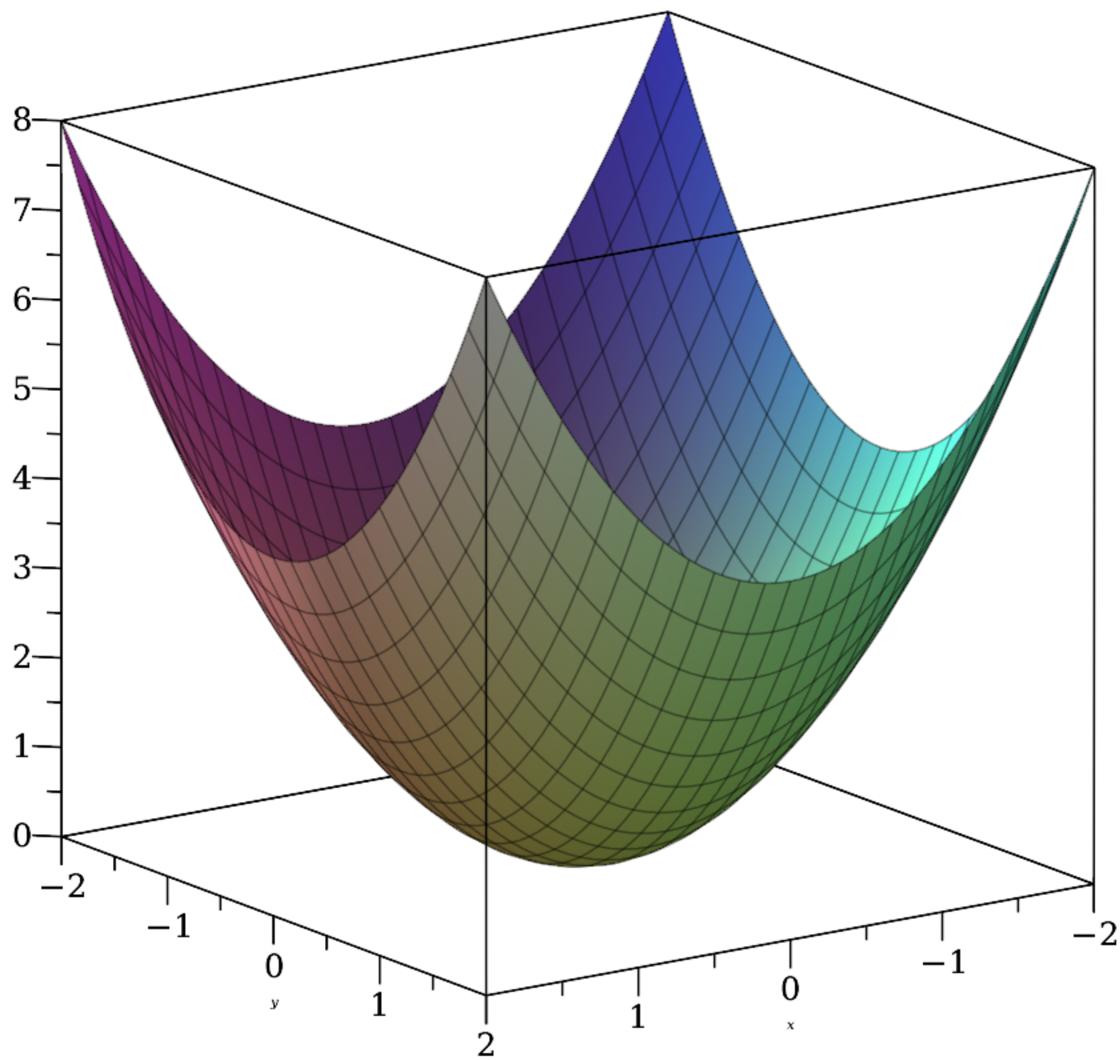

```
> # b)
> restart;
> with(plots);
```

[*animate*, *animate3d*, *animatecurve*, *arrow*, *changecoords*, *complexplot*, *complexplot3d*, *conformal*, *conformal3d*, *contourplot*, *contourplot3d*, *coordplot*, *coordplot3d*, *densityplot*, *display*, *dualaxisplot*, *fieldplot*, *fieldplot3d*, *gradplot*, *gradplot3d*, *implicitplot*, *implicitplot3d*, *inequal*, *interactive*, *interactiveparams*, *intersectplot*, *listcontplot*, *listcontplot3d*, *listdensityplot*, *listplot*, *listplot3d*, *loglogplot*, *logplot*, *matrixplot*, *multiple*, *odeplot*, *pareto*, *plotcompare*, *pointplot*, *pointplot3d*, *polarplot*, *polygonplot*, *polygonplot3d*, *polyhedra_supported*, *polyhedraplot*, *rootlocus*, *semilogplot*, *setcolors*, *setoptions*, *setoptions3d*, *shadebetween*, *spacecurve*, *sparsematrixplot*, *surfdata*, *textplot*, *textplot3d*, *tubeplot*] **(67)**

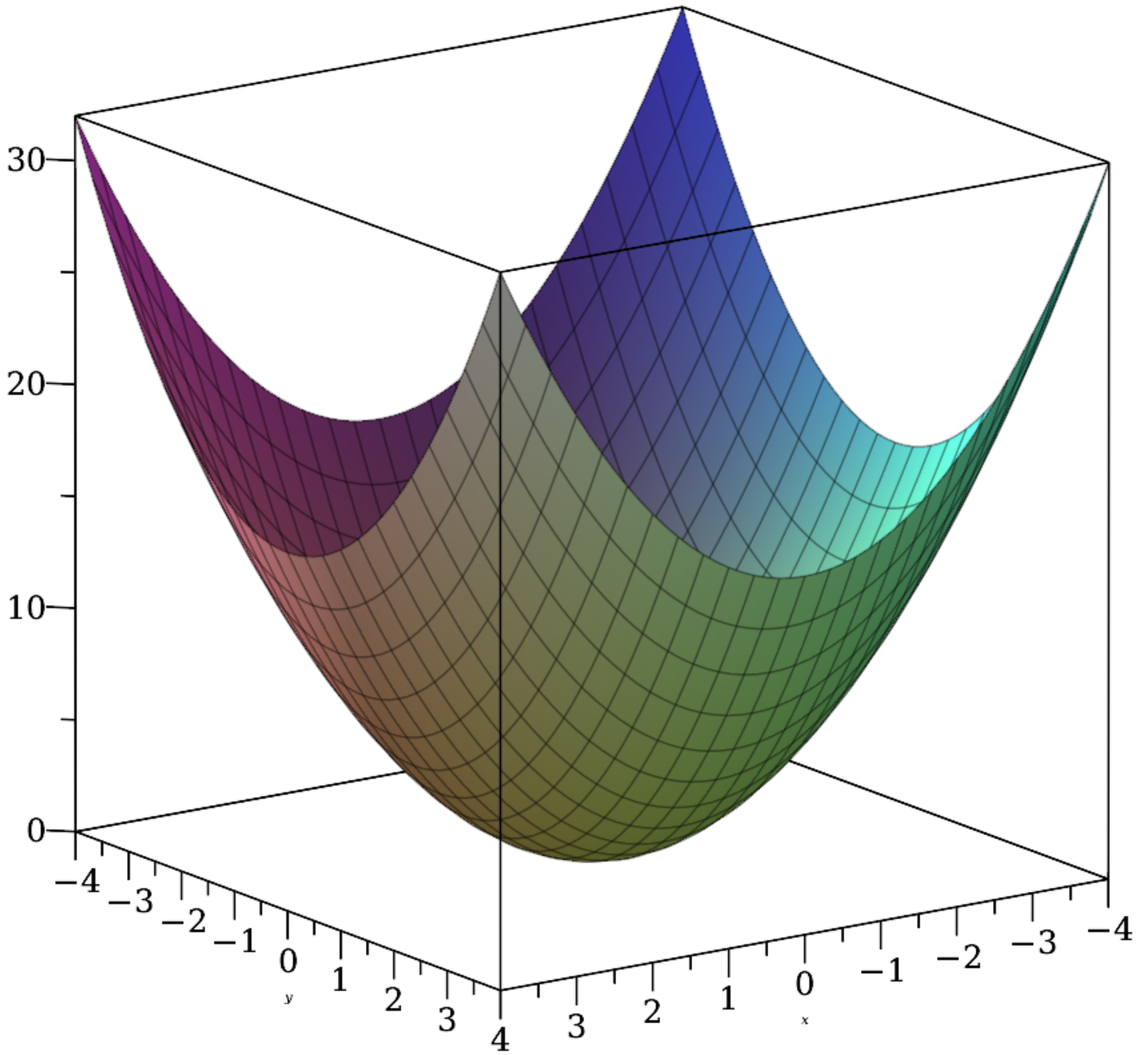
```
> implicitplot(y^3+y^2-5*y-x^2=-4,x=-6..6,y=-4..4, grid=[300,300],
  scaling=constrained);
```



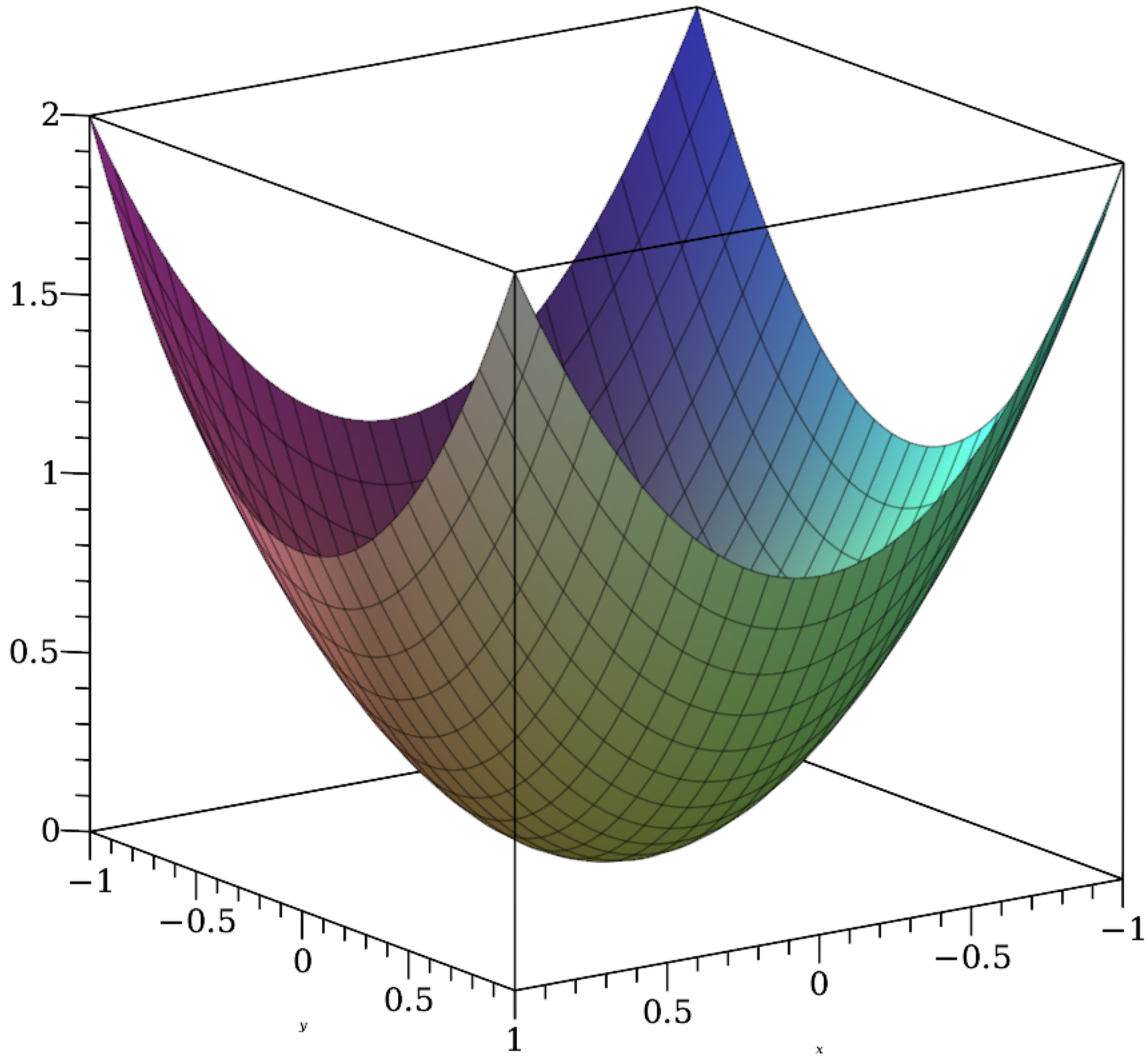

```
> restart;
> plot3d(x^2+y^2,x=-2..2,y=-2..2,axes=boxed);
```




```
> plot3d(x^2+y^2,x=-4..4,y=-4..4,axes=boxed);
```




```
> plot3d(x^2+y^2,x=-1..1,y=-1..1,axes=boxed);
```




```
> with(plots);
```

[*animate*, *animate3d*, *animatecurve*, *arrow*, *changecoords*, *complexplot*, *complexplot3d*, *conformal*, *conformal3d*, *contourplot*, *contourplot3d*, *coordplot*, *coordplot3d*, *densityplot*, *display*, *dualaxisplot*, *fieldplot*, *fieldplot3d*, *gradplot*, *gradplot3d*, *implicitplot*, *implicitplot3d*, *inequal*, *interactive*, *interactiveparams*, *intersectplot*, *listcontplot*, *listcontplot3d*, *listdensityplot*, *listplot*, *listplot3d*, *loglogplot*, *logplot*, *matrixplot*, *multiple*, *odeplot*, *pareto*, *plotcompare*, *pointplot*, *pointplot3d*, *polarplot*, **(68)** *polygonplot, polygonplot3d, polyhedra_supported, polyhedraplot, rootlocus, semilogplot, setcolors, setoptions, setoptions3d, shadebetween, spacecurve, sparsematrixplot, surfdata, textplot, textplot3d, tubeplot*]

```
> contourplot(x^2+y^2,x=-3..3,y=-3..3,contours=[1,4,9],scaling=
  constrained);
```